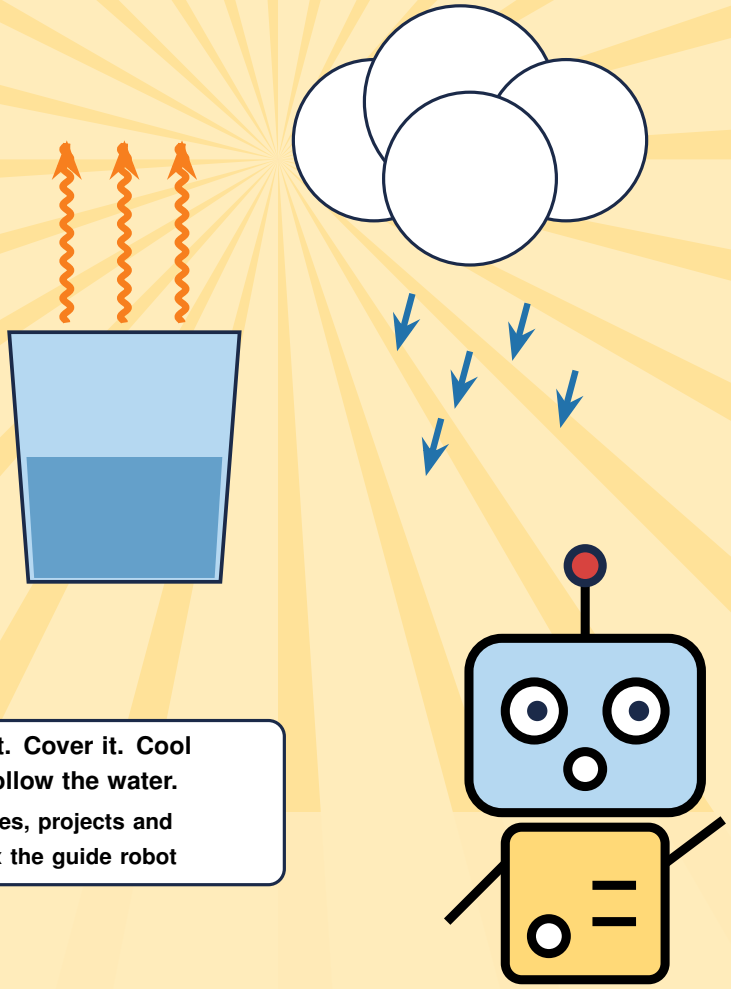
SCIENCE
Book 5

Where the Water Goes



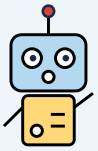
Mark it. Cover it. Cool it. Follow the water.

Puzzles, projects and Multix the guide robot

Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

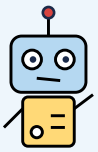


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop. Everything is at home or at school.
- **What did we keep?** After you simplify something, such as keeping the same conditions in a test or averaging several throws, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them, and Book 1 introduced them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 45.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I left a cup of water on a shelf and the next day there was less. I decided the water had vanished. Chapter 1 will explain why I should be embarrassed.

In this book

1	A Cup That Seems to Lose Water	5
2	Drying Up	10
3	Drops on a Cold Glass	16
4	A Cup With a Lid	21
5	The Long Loop	26
6	What the Diagram Leaves Out	32
⋆	Slop check	38
⋆	Answers	45
⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school. Measuring comes from Book 1 and fair testing from Book 2, and both come back here.

- two clear cups, alike
- a clear jar with a lid
- a ruler marked in mm
- a spoon
- a marker pen
- a few ice cubes
- food colouring or ink
- cling film and tape
- a zip bag
- paper towel
- a small plant or moss

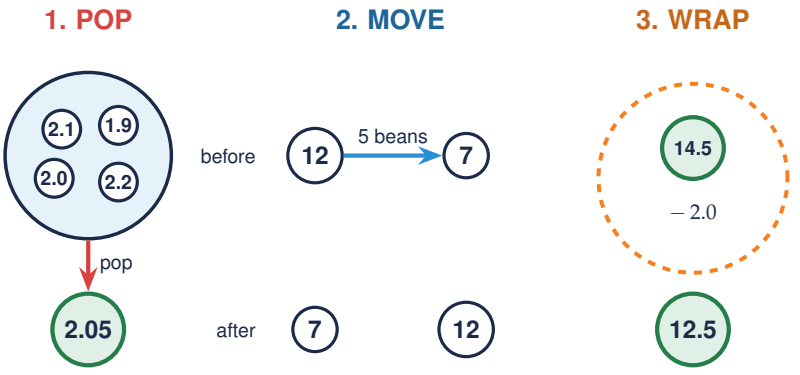
■ **Adult help:** Use cold tap water and ice, never hot water from a kettle or stove. Wipe up spills so nobody slips. Do not drink water that has been in an experiment. Nothing here needs a flame or mains electricity.

Units in this book

We use the metric system: centimetres (cm), metres (m) and seconds (s). For water we use millimetres of height in a cup and spoonfuls, so one spoonful is a small, countable amount. A spoonful is about 5 millilitres (5 mL). There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.

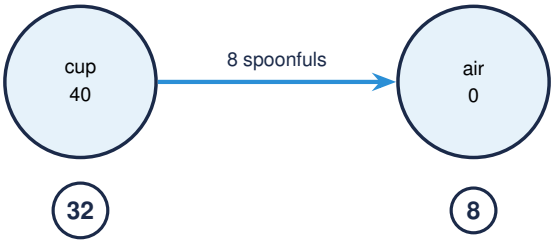


Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

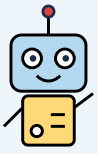
Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

Water moves between circles



The cup lost 8. The air gained 8. The total is still 40.



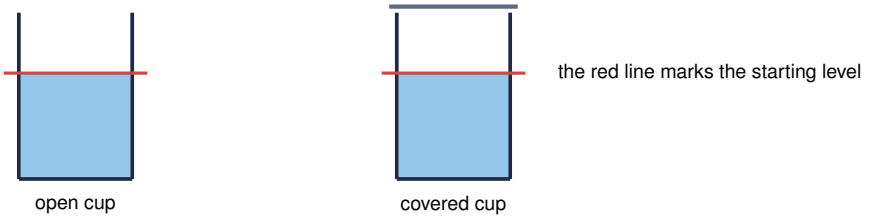
A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 A Cup That Seems to Lose Water

Marking a level and finding out where the water went

A puddle on a sunny path is gone by afternoon. A cup of water on a shelf has a little less water each day. Where did the water go?

There are three ideas. The water might have **soaked away** into something. It might have **stopped existing**. Or it might have **moved** somewhere we cannot easily see. A good test does not just pick the idea we like. It sets one idea against another and sees what happens.



Try it: Mark the level

You need: two clear cups that are the same, water, a marker pen, a ruler, cling film, tape.

1. Pour the same amount of water into both cups, about 6 cm deep. Use the ruler and pour until each is exactly the same.
2. Draw a line on the outside at the water level of each cup.
3. Cover **one** cup tightly with cling film and tape. Leave the other open.
4. Stand both on the same shelf, side by side, out of direct sun. Measure the water depth in millimetres at the same time each day for five days.

Day	0	1	2	3	4	5
Open cup (mm)						
Covered cup (mm)						

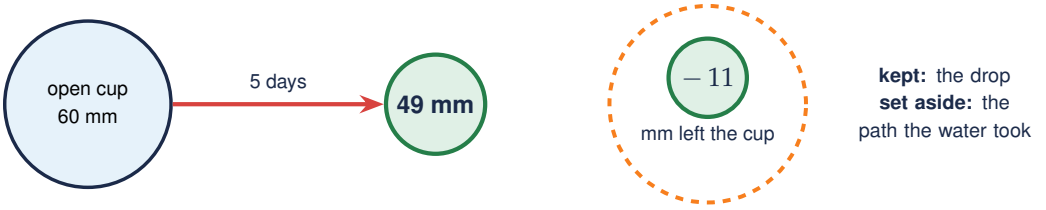
■ **Adult help:** Use cold tap water. Put the cups where they will not be knocked over. Do not drink the water afterwards.

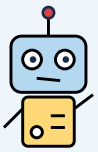
Reading a level honestly

Water curves a little where it touches the glass. Read the flat middle of the surface, with your eye level with it, and round to the nearest millimetre. A one-millimetre wobble in a reading is not a change in the water. We saw why in Book 1.

Practice numbers, not from a real test

Open cup, in mm: 60, 58, 56, 53, 51, 49. Covered cup: 60, 60, 60, 59, 60, 60.
The open cup dropped $60 - 49 = 11$ mm in 5 days. That is $11 \div 5 = 2.2$ mm a day, or about 2 mm.
The covered cup read 59 once and 60 at the next reading, so the 59 is a wobble, not a loss.





The open cup dropped 11 mm and the covered cup did not, so the extra water just stopped existing. I will write that in my notebook. Should I?

What makes this correction trustworthy?

What we checked. We changed one thing, the cling film, and kept the cups, the water, the shelf and the start level the same, then measured both cups every day.

What this check gives us reason to trust. The covered cup held its level and the open one dropped steadily, so the lid made the difference.

What it does not show. Where the missing water went, or whether the same drop would happen in a cold room or in the sun. We tested one shelf.

What did we keep?

We turned a cup of water into a number of millimetres. **We kept** _____ **and we set aside** _____.

Puzzle 1.1: Which cup lost water?

Open cup: 40, 38, 37, 35 mm. Covered cup: 40, 40, 40, 40 mm.
How much did the open cup drop? _____ mm The covered cup? _____ mm

Puzzle 1.2: Water a day

An open cup read 50, 47, 45, 42, 40 mm over four days after the first reading.
Total drop: _____ mm Average per day: _____ mm

Puzzle 1.3: Predict the level

At 2 mm a day, a cup that starts at 50 mm will read what after 10 days? _____ mm
Why is that only a guess? _____

Puzzle 1.4: Wobble or loss?

A covered cup read 60, 60, 59, 60, 60 mm. Did it lose water? _____
What makes you think so? _____

Puzzle 1.5: Spot the unfair test

Ana put her open cup on a sunny window sill and her covered cup in a dark cupboard. The open cup lost water. She says the lid saved it. What is wrong?
_____

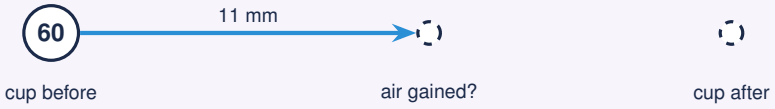
Puzzle 1.6: Three ideas

The water might have *soaked away*, *stopped existing* or *moved*. The covered cup kept all its water. Which idea does that make less likely for the open cup, and why?

_____

Puzzle 1.7: Draw the move

The open cup started at 60 mm and ended at 49 mm. Fill in the empty circles.



Puzzle 1.8: A puddle

One puddle in the sun is gone by noon. A puddle in the shade of a wall is still there at night. What is different about the two puddles? _____

Puzzle 1.9: Around you

Name two places at home where water seems to go away by itself.

_____

_____

Puzzle 1.10: Missing water

A student says "the water in the open cup went through the glass". If that were true, what would happen to the covered cup? _____

Cup diary

Keep your own record. Fill in what you see each day, and draw the level on the cups.

Day	Open cup (mm)	Covered cup (mm)	What I saw on the cling film
0			
1			
2			
3			
4			
5			

Do you see anything on the inside of the cling film? Draw it and write about it. You will see why it matters in Chapter 3.

2 Drying Up

Water leaving as a gas you cannot see

Wet washing dries on a line. A wet towel dries on a hook. Nobody sees the water go. It leaves as **water vapour**, which is water as an invisible gas mixed into the air. Water turning into vapour is called **evaporation**.

The covered cup in Chapter 1 kept its water because the cling film kept the vapour in. The open cup let the vapour drift away.

Vapour is invisible. Steam and mist are not vapour.
The white cloud above a kettle, and a mist over a pond, are made of tiny drops of liquid water. Real vapour has no colour and cannot be seen. You will meet the tiny drops again in Chapter 3.

Try it: A drying race

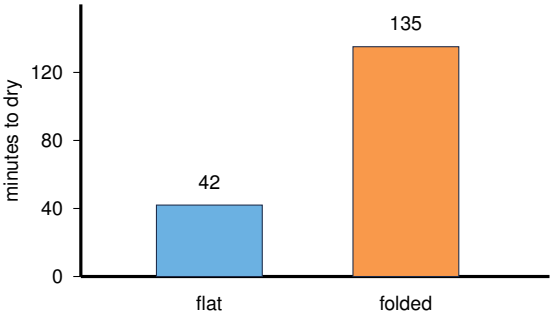
You need: two paper towel squares the same size, two plates, a spoon, a clock.

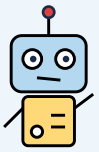
1. Put 4 spoonfuls of water on each square, so they are equally wet.
2. Lay one square **flat** on a plate. Fold the other into four and put it on a plate.
3. Put both plates side by side in the same spot. Check every 10 minutes. A square is dry when it feels dry and no dark patch is left.
4. Write the number of minutes each took. Do the race three times.

	Race 1	Race 2	Race 3	Average
Flat (min)				
Folded (min)				

Practice numbers, not from a real test

Flat: 40, 44, 42 minutes. Total 126, average $126 \div 3 = 42$, spread 4.
Folded: 130, 140, 135 minutes. Total 405, average $405 \div 3 = 135$, spread 10.
The folded square took $135 \div 42 \approx 3.2$, so about three times as long. The gap between the averages is much bigger than either spread.





Water only evaporates when it is hot. On a cold day my towel will never dry. I am very sure about this. Should I be?

What makes this correction trustworthy?

What we checked. We wet each square with the same amount, used the same paper, the same spot and the same clock, and repeated the race three times.

What this check gives us reason to trust. The three flat times were all near 42 and the three folded times all near 135, so the difference is far larger than the wobble between repeats.

What it does not show. Whether it holds for cloth, for other rooms, or for water that is not spread over paper. We changed only how the paper lay.

What did we keep?

We reduced drying to a number of minutes. **We kept** _____ **and we set aside** _____.

Warm and cool

Is evaporation faster in the warm? Test it with the *same* kind of towel square, the same amount of water and the same way of lying. Change only the place.

Try it: Sun or shade?

Put one flat wet square on a plate on a sunny window sill and one on a plate on a shady shelf. Check every 10 minutes. Do it three times.

	Try 1	Try 2	Try 3	Average
Sunny (min)				
Shady (min)				

Practice numbers

Sunny: 54, 60, 57 minutes. Total 171, average 57, spread 6.
Shady: 96, 102, 99 minutes. Total 297, average 99, spread 6.
The shady square took $99 - 57 = 42$ minutes longer.

What makes it quicker?

Change	Evaporation is usually quicker when
Warmth	the water and air are warmer
Spread	the water is spread thin over a bigger area
Moving air	a breeze carries the vapour away
Dry air	the air already holds little vapour

Notice that one fair test changes one of these at a time. Folding the square changed the spread. Sun against shade changed the warmth.

What makes this correction trustworthy?

What we checked. We repeated each placement three times and compared the averages.

What this check gives us reason to trust. The sunny times were all below the shady ones, by far more than the spread of six minutes.

What it does not show. That warmth was the only cause. A sunny sill may also be breezier or drier than a shelf.

What did we keep?

We said the sun makes things dry faster. **We kept** _____ **and we set aside** _____.

Add a breeze

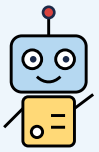
Try it: Still air or fanned?

Dry two flat wet squares the same way as before, one in still air and one with a book or card waved gently over it from a short way off. Keep the plates and place the same. Do it three times.

	Try 1	Try 2	Try 3	Average
Still air (min)				
Fanned (min)				

Practice numbers

Still air: 42, 40, 44 minutes. Total 126, average 42, spread 4.
Fanned: 24, 28, 26 minutes. Total 78, average 26, spread 4.
The fanned square dried $42 - 26 = 16$ minutes sooner. A breeze carries vapour away, so more can leave.



That means washing dries well on a windy day, even a cool one. Now my wet towel and I have something to talk about.

Puzzle 2.1: Average the races

Flat squares dried in 36, 40 and 38 minutes. Folded squares dried in 120, 126 and 132 minutes.
Flat average: _____ Folded average: _____

Puzzle 2.2: Which dries first?

Which of these dries first? Write 1 for first and 3 for last.
A towel on a line in a breeze _____ the same towel in a heap on the floor _____ the same towel folded in a closed drawer _____

Puzzle 2.3: Spot the unfair test

Sam dried a small square in the sun and a big square in the shade. The small one dried first. He says sun makes drying faster. List two things that changed. _____

Puzzle 2.4: How many times as long?

The folded square took 135 minutes and the flat one 42 minutes. About how many times as long did the folded square take? Round to a whole number. _____

Puzzle 2.5: Vapour or drops?

Write V if it is vapour and D if it is tiny drops.
(a) The cloud above a kettle _____ (b) The water in dry air that you cannot see _____
(c) Mist on a pond _____

Puzzle 2.6: Wash day

Washing dries on a cold, windy day more slowly than on a warm, windy one. It still dries. What does that tell you about Multix's claim? _____

Puzzle 2.7: Move the water

A towel holds 4 spoonfuls of water. After it is dry, where is the 4 spoonfuls? Fill in the circles.



Puzzle 2.8: Why folded is slower

Say in one sentence why a folded towel dries more slowly than a flat one. _____

Puzzle 2.9: Spread out

Which dries faster: a cup of water, or the same water in a wide tray? _____
Why? _____

Puzzle 2.10: Make a test

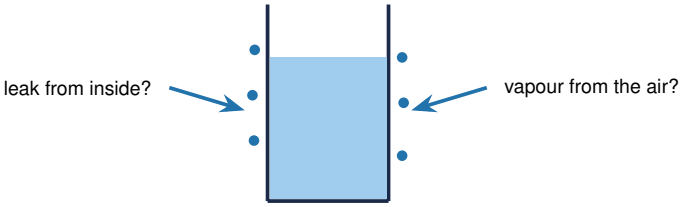
Plan a fair test of one thing that might change how fast a towel dries. Change: _____ Keep the same: _____ Measure: _____

3 Drops on a Cold Glass

Finding where the water came from

Fill a glass with ice water on a warm day and wait. Little drops appear on the *outside* of the glass. They grow, join, and run down. Where did that water come from?

There are two easy ideas. The water might have **leaked through the glass** from the inside. Or it might have come **out of the air** around the glass. Both could be true. A test can tell them apart.



Try it: Cold glass and colour

You need: two glasses that are the same, cold water, ice, food colouring or a drop of ink, a clock, paper towel.

1. Dry the outside of both glasses. Fill one with ice water and add a few drops of food colouring. Fill the other with room-temperature water and the same colouring.
2. Set them side by side. Count the seconds until you first see drops on the outside. Stop after 10 minutes if there are none.
3. Dab a drop with a white paper towel. Is it coloured or clear?
4. Do it three times.

	Try 1	Try 2	Try 3	Average
Cold glass: seconds to first drops				
Room glass: seconds to first drops				
Colour of drops (clear/coloured)				

■ **Adult help:** Food colouring stains clothes and hands. Use cold water and ice only.

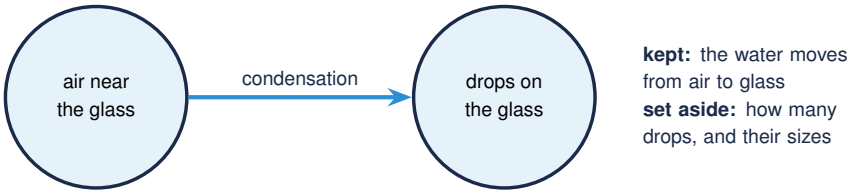
Practice numbers, not from a real test

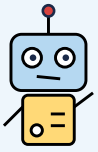
Cold glass: 90, 80, 100 seconds. Total 270, average 270 ÷ 3 = 90 seconds, spread 20.
Room glass: no drops in 10 minutes, all three tries.
The drops from the cold glass were clear every time.

What the test says

If the glass leaked, the drops would be coloured like the water inside. They were clear. So the drops did not come from the inside. The air was the likely source. The room glass fits that too, since without a cold surface no drops formed.

Air always holds some invisible water vapour. When air touches something cold, the air next to it cools. Cool air can hold less vapour, so some vapour turns back into tiny drops of liquid water. That change, vapour to liquid, is **condensation**. It is evaporation in reverse.





The drops are on the outside of the glass and the water is on the inside, so the glass must be leaking a tiny bit. I am a careful robot. Did I miss a test?

What makes this correction trustworthy?

What we checked. We repeated the cold glass three times and added colour to the water inside to see whether the drops carried it.

What this check gives us reason to trust. The drops were clear every time, which is hard to explain if they came through the glass.

What it does not show. How much water came from the air, or whether a tiny, invisible leak also happened. The colour test only rules out a leak big enough to carry visible colour.

What did we keep?

We said the drops came from the air. **We kept** _____ **and we set aside** _____.

Fog on a spoon

Try it: Breath on a cold spoon

Put one metal spoon in the fridge for 10 minutes and leave another at room temperature. Breathe gently on each. Which one fogs? _____

Does the fog stay? What happens as the spoon warms? _____

Practice result

The cold spoon fogged at once and cleared in about 10 seconds as it warmed. The room spoon barely fogged. Your breath carries extra vapour, and the cold spoon cools it enough for drops to form.

The cling film again

Look back at the covered cup in Chapter 1. Drops on the inside of the film are condensation too. In a covered cup the vapour has nowhere to go, so it meets the film and turns back to drops. Some run back into the water.

Puzzle 3.1: Average the times

Cold glass: 90, 80 and 100 seconds. Total _____ Average _____ s

Puzzle 3.2: Colour clue

Blue water inside a cold glass gave clear drops outside. Which idea does that support? Circle one.

leaked through **came from the air** **cannot tell**

Puzzle 3.3: No drops

Why did the room-temperature glass stay dry outside?

_____

Puzzle 3.4: Unfair test

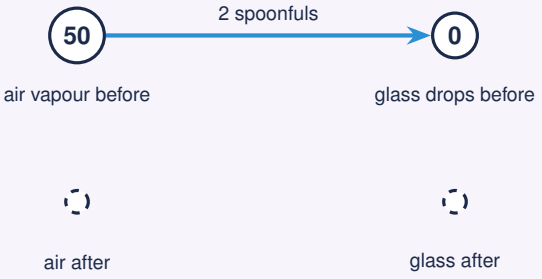
Mia put a cold glass in the kitchen and a warm glass outside in the sun. Only the cold glass had drops. Name two things that were different. _____

Puzzle 3.5: Cold things

Name two cold things that collect drops. _____

Puzzle 3.6: Move the water

Suppose a cold glass collected 2 spoonfuls of water from a room. Fill in the circles.



Puzzle 3.7: A sentence for the data

Mark S if the data supports it, G if it is a guess about why, and B if it is too big for the data.
(a) "The drops from the cold glass were clear." _____
(b) "The drops came out of the air." _____
(c) "All drops everywhere come from air." _____

Puzzle 3.8: Morning window

On a cold morning the inside of a window has drops. Is the glass cold or warm? _____ Where could the water come from? _____

Puzzle 3.9: Why the film?

In the covered cup, why might drops form on the cling film? _____

4 A Cup With a Lid

Keeping count of the water

In Chapter 1 an open cup lost water and a covered cup did not. That suggests the water went into the air, not into nothing. Here is a way to check, by keeping a **ledger**. A ledger is a list of where the amounts are, written so the total can be added up.

Try it: A water ledger

You need: two jars that are the same, one with a lid, a spoon, water, a marker pen.

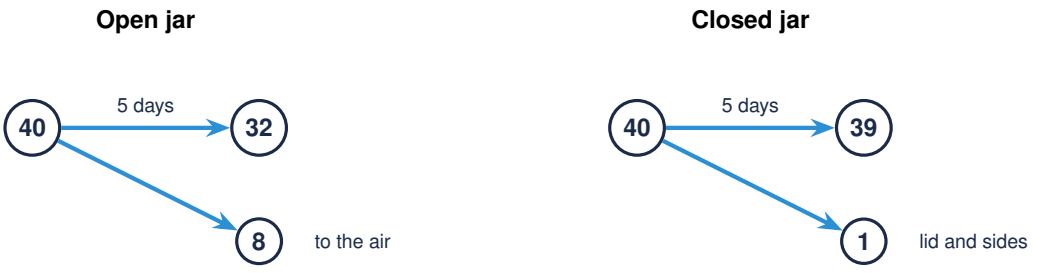
1. Put exactly 40 spoonfuls of water in each jar. Count each spoonful. Mark the level on the outside.
2. Close one jar tightly with its lid. Leave the other open. Put them side by side for 5 days.
3. At the end, use the same spoon to take the water out of each jar and count the spoonfuls.
4. Look at the inside of the lid and the glass above the water. Note any drops. Write down the numbers.

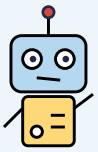
	Start	End: in the jar	Drops on lid and sides
Open jar	40		
Closed jar	40		

■ **Adult help:** Keep the jars on a steady shelf. Wipe up spills. The water is not for drinking afterwards.

Practice numbers, not from a real test

Open jar: 32 spoonfuls left. $40 - 32 = 8$ spoonfuls went into the air.
Closed jar: 39 spoonfuls left, plus about 1 spoonful worth of drops on the lid and sides. $39 + 1 = 40$.
The open jar's total only adds up if we count the air. The closed jar's total adds up without it.





My closed jar had 39 spoonfuls in it at the end, and I started with 40. So one spoonful was destroyed. I always trust my arithmetic. Should I trust this answer?

What makes this correction trustworthy?

What we checked. We counted every spoonful at the start and the end, in an open jar and a closed one, and looked for water outside the main pool.

What this check gives us reason to trust. The closed jar's amounts added up to the start total, and the open jar's loss matched what the air could have gained.

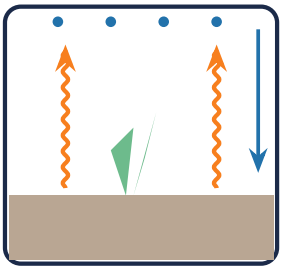
What it does not show. That nothing is ever lost. A spoon counts in whole steps, so a gap of one spoonful is about the size of the measuring error.

What did we keep?

We counted water in spoonfuls. **We kept** _____ **and we set aside** _____.

A tiny garden in a jar

A sealed jar with damp soil and a small plant, or some moss, is called a **terrarium**. The water cannot escape, so it goes round inside. It evaporates from the soil, condenses on the cool lid, and runs back down. The plant also lets water out of its leaves as vapour.



Try it: Make a terrarium

You need: a clear jar with a lid, a handful of soil, a small plant or some moss, a spoon, water.

Put soil in the jar, add the plant, sprinkle in 4 spoonfuls of water and close the lid. Mark the soil level. Leave it in a bright place, but not in hot sun. Look each day and draw what you see.

Day	1	2	3	4	5
Drops on the lid?					
Soil looks (dry/damp)					

Steps in order

Here are the steps in a closed jar, in order.

1. Water evaporates from the soil.
2. The vapour touches the cooler lid.
3. It condenses into drops.
4. The drops run down and soak back into the soil.

In a jar the loop is small. Chapter 5 follows the same steps on a much larger scale.

Puzzle 4.1: Do the ledger

An open jar started with 40 spoonfuls and ended with 31. How many went into the air? _____

Puzzle 4.2: Closed jar

A closed jar started with 40 spoonfuls. At the end it held 38, and about 2 spoonfuls' worth of drops were on the lid. Add it up. _____ Does it match the start? _____

Puzzle 4.3: Find the gap

A closed jar started with 40. It ended with 36 in the pool and 3 on the lid. Total: _____
What is the gap? _____
Could a gap this size be a counting error? _____ What would you do? _____

Puzzle 4.4: Order the steps

Number these from 1 to 4.
_____ Drops run down and soak in. _____ Water evaporates from soil.
_____ Drops form on the lid. _____ Vapour reaches the cooler lid.

Puzzle 4.5: Plant in a jar

A plant in a closed jar stayed alive for weeks without watering. Why could that happen? _____

Puzzle 4.6: Open or closed?

Jar A kept 40 spoonfuls for 5 days. Jar B dropped to 32. One was closed and one was open. Which was which? _____

Puzzle 4.7: Missing from the ledger

An open jar started with 40 and ended with 32. A student says 8 spoonfuls disappeared forever. Write a better sentence. _____

Puzzle 4.8: Draw the ledger

A cup holds 24 spoonfuls. After a week it holds 18. Fill in the ledger circles.



Puzzle 4.9: Sealed or open

Give one good reason a scientist would use a sealed jar to study where water goes. _____

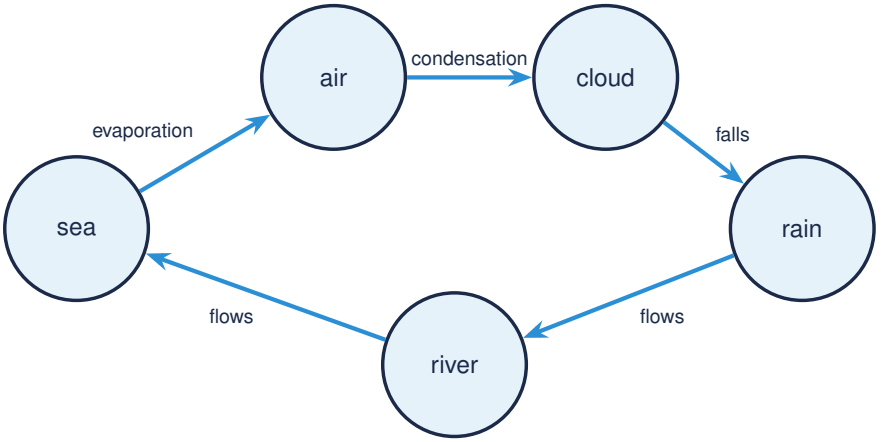
Puzzle 4.10: Your terrarium

After five days, did drops form on the lid? Draw what you saw. _____

5 The Long Loop

Following water from sea to sky and back

Evaporation, condensation and a place to flow back make a loop. On a huge scale, water evaporates from the sea and from wet land, rises as vapour, cools and condenses into tiny drops that make clouds, falls as rain or snow, and runs back toward the sea. That is the **water cycle**.



Try it: A solar still

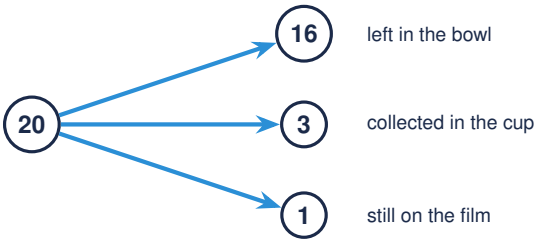
You need: a large bowl, a small cup, cling film, tape, a small stone, water, food colouring, a spoon, a sunny window sill.

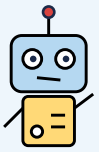
1. Put 20 spoonfuls of coloured water in the bowl. Stand the small cup in the middle. Do not let water get in the cup.
2. Cover the bowl with cling film, taped tight. Put the small stone on the film right over the cup so the film dips toward it.
3. Put the bowl in the sun for a day. Look for drops on the film. Where do they run to?
4. At the end, count spoonfuls in the cup, in the bowl, and note any drops left on the film. Is the cup's water coloured?

■ **Adult help:** Cold tap water only. Do not drink any water from this activity. Put the bowl where it will not be knocked.

Practice numbers, not from a real test

Start: 20 spoonfuls of coloured water.
End: 16 in the bowl, 3 in the cup, and about 1 spoonful worth still as drops on the film.
$16 + 3 + 1 = 20$. The water in the cup was clear. The colour stayed behind in the bowl.





The clear water in my cup is new. Clouds must make new water out of nothing. How else would rain appear? I like my idea. Do you?

What makes this correction trustworthy?

What we checked. We counted every spoonful at the start and the end, including the drops on the film, and we checked the cup's water for colour.

What this check gives us reason to trust. The amounts added up to the start total, and the clear water in the cup shows the water travelled as vapour and left the colour behind.

What it does not show. Whether the real sea and sky work exactly the same way. A bowl in a window is a small stand-in for something enormously bigger.

What did we keep?

We turned the bowl into a small water cycle. **We kept** _____ **and we set aside** _____.

Rain in a bag

Try it: A cloud in a bag

Put a spoonful of water in a zip bag, seal it, tape it to a sunny window and watch for a morning. Where do drops form? _____
What happens when drops get big? _____

Practice result

After two hours small drops lined the top of the bag, where it was cooler. By the afternoon bigger drops ran down to the bottom. The water went round a tiny loop, much like the water cycle.

What a diagram of the cycle leaves out

Every diagram of the water cycle is a model, a simple stand-in. Here is what the arrows above keep and what they set aside.

Part of the diagram	What it keeps	What it sets aside
Boxes for the stages	the order of the stages	how many different places each is
Arrows between boxes	the direction water moves	how long each step takes
One loop	the same water going round	plants, ice, underground water
Labelled steps	the names of the changes	the amounts of water in each

Water stays in the air for roughly a week or so on average. It can stay in the deep ocean or in ice for far longer. The diagram's arrows are all the same size, but the real steps are not.

What makes this correction trustworthy?

What we checked. We checked the model against the jar, the bowl and the bag, each of which keeps the total.

What this check gives us reason to trust. Each small test had the same order: liquid, vapour, drops, liquid again.

What it does not show. That a model with five boxes can predict rainfall, or how long water stays anywhere. It is a map of the order, not of the amounts or the times.

Puzzle 5.1: Do the ledger

A solar still started with 20 spoonfuls. At the end the bowl held 15 and the cup held 4.
How many spoonfuls are unaccounted for? _____
Where might they be? _____

Puzzle 5.2: Put them in order

Number the steps from 1 to 5.
_____ Rain falls. _____ Water evaporates from the sea. _____ Rivers flow to the sea.
_____ Vapour condenses into cloud drops. _____ Vapour rises into the air.

Puzzle 5.3: Which step?

Write E for evaporation, C for condensation or R for rain.
(a) A puddle shrinks _____ (b) Dew on grass _____ (c) A cloud forms _____
(d) A wet street dries _____ (e) A steamy mirror _____ (f) A storm _____

Puzzle 5.4: Clear water

The water in the cup of the solar still was clear, but the water in the bowl was coloured.
What does this tell you about what travelled? _____

Puzzle 5.5: Follow a drop

Start at the sea. Which of these routes is possible? Circle it.
sea → **air** → **cloud** → **rain** → **river** → **sea** **sea** → **cloud** → **river** → **air**

Puzzle 5.6: Rain or snow?

In cold weather the cloud drops may freeze. What might fall instead of rain? _____

Puzzle 5.7: Missing from the diagram

Name two things the five-box diagram of the water cycle leaves out. _____

Puzzle 5.8: Same size arrows

The diagram draws every arrow the same size. Why might that be misleading? _____

Puzzle 5.9: Closed loop

In the bag and the jar the water went round and round. Where in the real world is the water cycle *not* closed in a small way? _____

Puzzle 5.10: Your loop

Draw your own loop for a puddle in the sun. Use circles and arrows. _____

Draw your own water cycle

Draw a loop for a place you know, such as a lake, a garden or a rooftop. Use circles for the stages and arrows for the moves. Then fill in the three sentences.



My diagram keeps: _____

It sets aside: _____

A question it cannot answer: _____

6 What the Diagram Leaves Out

Asking what a picture of the cycle can and cannot tell us

A diagram of the water cycle is a model. It keeps the order of the stages and drops almost everything else: the places, the times, the amounts, the plants, the ice. Whether a model is good depends on the question you ask of it.

Model	What it keeps	What it drops
A level in a cup	how much water is there	where the missing water went
A ledger	the total across places	the path between them
A colour test	whether drops came from inside	tiny leaks that carry no colour
A solar still	the order of the changes	real sea, real sky, real time
A cycle diagram	the order of the stages	times, places, amounts

A question for every model

Ask of any diagram: what question is it for? Then ask: what would it get wrong if I used it for a different question?

Try it: A rain gauge

You need: a straight-sided clear cup or jar, a ruler marked in mm, a spot outdoors away from trees.

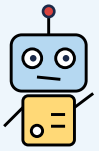
1. Stand the cup outside before rain is due, level on the ground or a flat shelf.
2. Each day at the same time, measure the depth of water in mm. Then empty the cup.
3. Write the readings for a week.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Rain (mm)							

Practice numbers, not from a real week

Mon 4, Tue 0, Wed 7, Thu 2, Fri 0: total $4 + 0 + 7 + 2 + 0 = 13$ mm in five days.
The average is $13 \div 5 = 2.6$, about 3 mm a day.

Each 1 mm of rain is 1 litre of water on every square metre of ground. So 13 mm is 13 litres on each square metre.



The diagram has five boxes and five arrows, so it must be a perfect picture of the real sky. I trust it for everything. Am I being careful?

What makes this correction trustworthy?

What we checked. We asked what question the diagram is meant to answer, then checked it against real measurements, a jar, a bowl and a rain gauge.

What this check gives us reason to trust. The diagram gets the order of the stages right, and the small tests kept their totals.

What it does not show. Anything about how long water stays, how much falls in a place, or how much goes where. For those questions we need other models, such as a ledger with numbers.

What did we keep?

We turned a week of rain into one number. **We kept** _____ **and we set aside** _____.

Puzzle 6.1: Rain total

Rain gauge readings: 5, 0, 3, 6, 1 mm. Total _____ mm Average per day _____ mm

Puzzle 6.2: Which question?

Mark Y if the five-box diagram can answer the question and N if it cannot.
(a) What stage comes after the cloud? _____
(b) How long does a drop stay in the air? _____
(c) How much rain falls on my town? _____

Puzzle 6.3: Improve the diagram

Name one thing you would add to the diagram to make it answer a new question, and the question. _____

Puzzle 6.4: Weather clue

Match each clue to a stage. E (evaporation), C (condensation) or R (rain falling).
(a) Washing dries on a line _____ (b) A cold can gets wet outside _____
(c) A puddle appears under grey clouds _____

Puzzle 6.5: Wrap the claim

A student writes “the water in the open cup vanished”. Use what you know to wrap that claim with a better one. _____

Puzzle 6.6: Ledger check

A cup started with 30 spoonfuls and ended with 22. How many moved into the air? _____

Puzzle 6.7: Unfair rain test

Ben put one rain gauge under a tree and another in an open field. They read 2 mm and 9 mm. He says the field gets more rain. What is wrong with the test? _____

Puzzle 6.8: Model or real thing?

Write M for model and R for real thing.
(a) A drawing of the cycle _____ (b) A cloud _____ (c) A closed jar with a plant _____

Puzzle 6.9: One question

Write a question that a ledger can answer but a five-box diagram cannot. _____

Puzzle 6.10: Rain again

8, 0, 0, 4, 3 mm. Total _____ mm

Puzzle 6.11: Missing step

Water went from the sea to a cloud. Name the step in between. _____

Puzzle 6.12: A cold morning

The outside of a cold bottle gets wet in a warm kitchen. Name the change. _____

Puzzle 6.13: Trust

A student says her jar kept all its water, so water is never lost anywhere. What should she check? _____

Puzzle 6.14: Draw the move: the lake

A lake loses 6 units to the air on a sunny day. Draw a move from the lake to the air and write the amount. Then write what the lake and the air hold now if they held 100 and 20 at the start.

Puzzle 6.15: Draw the loop: the bag

Draw the cloud-in-a-bag loop with circles: water, vapour, drops. Label the steps.

Puzzle 6.16: Draw the wrap

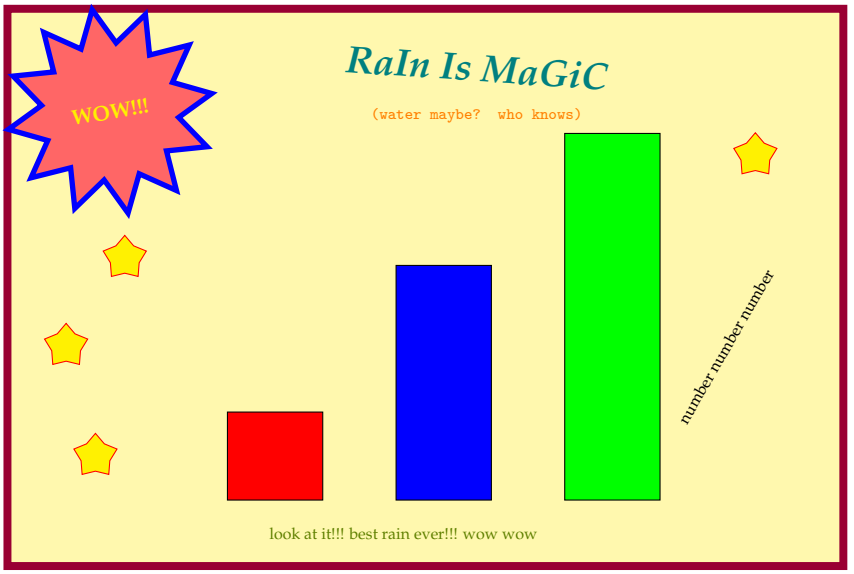
Draw a circle around “water vanished” and write the corrected claim outside it.

Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is whether the work shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty poster about rain. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so the bars cannot be read.

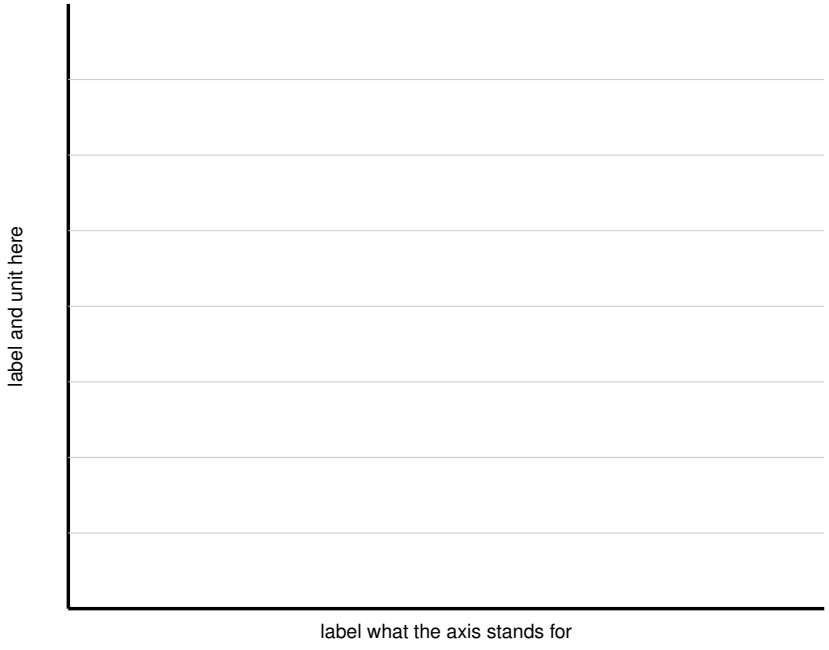
_____

(d) Which part is not relevant to rain at all?

_____

Now you try

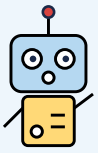
On a separate sheet, redraw a clear bar chart of rain each day: Mon 4 mm, Tue 0 mm, Wed 7 mm, Thu 2 mm, Fri 0 mm. Give it a title, label the axes with units, use one or two calm colours, and delete everything that does not help. Check it against the three words: salient, simple, relevant.



What did we keep, across the whole book?

For each chapter, say one thing that was simplified, what was kept, and what was set aside. A sample is done for you.

Chapter	What we simplified	What we kept	What we set aside
1 Level in a cup	the water as a height in mm	how much is left	where it went
2 Drying up	drying as minutes	which dries sooner	the path of the vapour
3 Cold glass	drops as one number	whether they were clear	the size of each drop
4 A cup with a lid	water as spoonfuls	the total	the exact path
5 The long loop	the cycle as boxes	the order	times and amounts
6 The diagram			



Everything you drew in this book was a simplification of something. A model is allowed to drop things, as long as you say which. Next time a puddle shrinks, you will know where it went.

Write a water report

1. **What I looked at.**
2. **What I changed, measured and kept the same.**
3. **My readings.** All of them.
4. **My answer.** With averages and spread.
5. **What I kept and what it hides.**

Pick one test from this book and write a report on it. Use the lines below.

Tests to try

Pick one and design it with the habits from this book: change one thing, keep the rest the same, repeat, and average.

1. Does the wind make a wet towel dry faster? Try a fan or a breezy window.
2. Does the shape of a dish change how fast water evaporates?
3. Does salty water evaporate faster or slower than fresh water?
4. How long does it take a cold can to form drops in different rooms?
5. Does a plant in a closed jar keep the jar's water for longer than an open one?
6. How much rain falls on different spots around your home?

Write the test you pick:

_____

_____

_____

_____

_____

_____

Change: _____ Measure: _____ Keep the same: _____

A week of water

Each day, notice one place water goes in or out of the air and say which stage it is.

Day	What I saw	Which stage
Mon		
Tue		
Wed		
Thu		
Fri		
Sat		
Sun		

Questions I still have

Write what you still wonder about water. Then pick one and think how to test it.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on vanishing. No. The covered cup kept its water, so the water in the open cup most likely moved into the air. **Keep box.** Kept: how much water was left. Set aside: the path the water took.

1.1 $40 - 35 = 5$ mm. 0 mm.
1.2 Drops of 3, 2, 3 and 2 mm, a total of 10 mm in four days, so $10 \div 4 = 2.5$ mm a day.
1.3 $10 \times 2 = 20$, so $50 - 20 = 30$ mm. It is a guess because the rate may change with weather.
1.4 No. The reading went to 59 and back to 60, so it is a wobble.
1.5 Two things changed, the lid and the place (sun against dark).
1.6 It makes "stopped existing" and "soaked away" less likely, since the covered cup in the same glass kept all its water. The water moved into the air.
1.7 Air gained 11 and the cup now holds 49.
1.8 The sun, which warms the water, and perhaps a breeze. The shade is cooler.
1.9 Answers vary, such as wet washing, a wet towel or a puddle.
1.10 The covered cup would lose water too. It did not.

Chapter 2

Multix on cold days. No. Evaporation is faster in the warm, but it also happens in the cold. Washing dries slowly on a cold windy day. **Keep box.** Kept: which dried sooner. Set aside: the path of the vapour.

2.1 Flat: $36 + 40 + 38 = 114$, average 38. Folded: $120 + 126 + 132 = 378$, average 126.
2.2 Line in a breeze 1, heap 2, folded in a closed drawer 3.
2.3 The size of the square and the place (sun against shade).
2.4 $135 \div 42 \approx 3.2$, so about 3 times.
2.5 (a) D (b) V (c) D.
2.6 Washing still dries on a cold day, so evaporation is not only for hot weather.
2.7 The air gained 4 spoonfuls.
2.8 In a fold less of the wet paper touches open air, so the vapour escapes more slowly.
2.9 The wide tray. The water is spread over more area.
2.10 Answers vary. Check that only one thing is changed.

Chapter 3

Multix on the leak. It missed the colour test. Coloured water inside gave clear drops outside, which points to the air. **Keep box.** Kept: water moved from the air to the glass. Set aside: the number and size of the drops.

3.1 $90 + 80 + 100 = 270$, and $270 \div 3 = 90$ seconds.
3.2 Came from the air.
3.3 It was not cold enough to cool the air next to it, so no vapour turned into drops.
3.4 The temperature of the glass and the place (kitchen against outside in the sun).
3.5 Examples: a cold can, a window on a cold morning, a cold spoon.
3.6 Air after 48, glass after 2.
3.7 (a) S (b) G (c) B.
3.8 The window is cold. The vapour comes from the warm air in the room.
3.9 The film is cooler than the vapour, so the vapour turns into drops on it.

Chapter 4

Multix on the missing spoonful. Not yet. A gap of one spoonful is about the size of the counting error, and some water is on the lid and sides. **Keep box.** Kept: the total. Set aside: the exact path.

4.1 $40 - 31 = 9$.
4.2 $38 + 2 = 40$. Yes.
4.3 $36 + 3 = 39$, so the gap is 1. It could be a counting error. Recount or repeat.

4.4 Drops run down 4, water evaporates 1, drops form on the lid 3, vapour reaches the lid 2.
4.5 The water goes round inside the jar, so the soil is watered again.
4.6 Jar A was closed and Jar B was open.
4.7 "8 spoonfuls moved into the air as vapour."
4.8 18 stays, and the air gained 6.
4.9 A sealed jar keeps the vapour in, so the total can be counted.
4.10 Answers vary.

Chapter 5

Multix on new water. No. The clear water in the cup came from the bowl by evaporation and condensation. **Keep box.** Kept: the order of the stages. Set aside: sizes, places and times.

5.1 $20 - 15 - 4 = 1$. It is probably drops on the film or the sides.
5.2 Evaporates 1, vapour rises 2, condenses into cloud drops 3, rain falls 4, rivers flow to the sea 5.
5.3 (a) E (b) C (c) R (d) E (e) C (f) R.
5.4 The water travelled as vapour. The colour stayed behind.
5.5 The first route.
5.6 Snow or hail.
5.7 Examples: plants, ice, underground water, the amounts, the times.
5.8 The real steps are not the same size or speed.
5.9 Answers vary. Water can stay in the deep ocean or in ice for a very long time.
5.10 Puddle to air to cloud to rain and back to the puddle.

Chapter 6

Multix on the diagram. No. A five-box diagram keeps the order but not amounts or times. **Keep box.** Kept: the total rain. Set aside: which day it fell.

6.1 $5 + 0 + 3 + 6 + 1 = 15$ mm, average 3 mm.
6.2 (a) Y (b) N (c) N.
6.3 Answers vary, for example numbers on the arrows to say how much.
6.4 (a) E (b) C (c) R.
6.5 "The water in the open cup moved into the air."
6.6 $30 - 22 = 8$.
6.7 Trees catch the rain, so two things changed.
6.8 (a) M (b) R (c) M.
6.9 Example: how much water is left in a cup after five days?
6.10 $8 + 0 + 0 + 4 + 3 = 15$ mm.
6.11 Evaporation, then condensation.
6.12 Condensation.
6.13 One jar does not show water is never lost anywhere. Check other jars and conditions.
6.14 The lake holds 94 and the air 26.
6.15, 6.16 Answers vary. Check that water moves and nothing vanishes.
Slop check. (a) You cannot tell: no title, no labels. (b) Many colours, stars, joke text. (c) A title, the days and the rain amounts with units. (d) The stars and jokes.

Words

condensation vapour turning back into drops of liquid.

evaporation liquid turning into vapour.

ledger a list of where the amounts are, so a total can be added up.

model a simple stand-in that keeps some things and drops others.

terrarium a sealed jar with soil and a plant.

vapour water as an invisible gas in the air.

water cycle the loop of water from sea to air to rain and back.

spread how far apart repeated readings are.

Notes for grown-ups

This book teaches that a diagram of the water cycle is a model: it keeps the order of the stages and drops the times, places and amounts. Each chapter has one activity, one worked example with made-up practice numbers, one trust box, one "what did we keep?" prompt and a page of puzzles.

Materials and safety. Clear cups and jars, a ruler, cling film, ice, food colouring, a zip bag. Use cold tap water only. Do not drink water from an experiment. Clean up spills.

Vapour and steam. Water vapour is invisible. Mist, steam clouds and clouds in the sky are tiny drops of liquid water.

The practice numbers were made up so the arithmetic is easy to check. Real results will vary with the room, the day and the equipment.

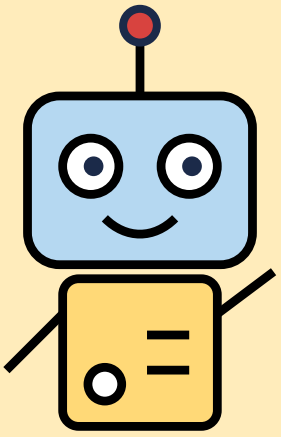
Sources

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- C. D. Ahrens. *Meteorology Today*. Cengage. For condensation, dew and clouds.
- J. R. Taylor. *An Introduction to Error Analysis*, 2nd edition. University Science Books, 1997. For repeats and spread.

Next in Science

Mixed Up and Sorted Out

A spoonful of salt goes into water and seems to vanish. Book 6 shows how to separate mixtures, and how to check that nothing was lost.



Ages 9 to 11 ⋆ Science Book 5